Decoding of Individuated Finger Movements Using Surface Electromyography

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***Abstract*—Upper limb prostheses are increasingly resembling the limbs they seek to replace in both form and functionality, including the design and development of multifingered hands and wrists. Hence, it becomes necessary to control large numbers of degrees of freedom (DOFs), required for individuated finger movements, preferably using noninvasive signals. While existing control paradigms are typically used to drive a single-DOF hook-based configurations, dexterous tasks such as individual finger movements would require more elaborate control schemes. We show that it is possible to decode individual flexion and extension movements of each finger (ten movements) with greater than *90% accuracy in a transradial amputee using only noninvasive surface myoelectric signals*. Further, comparison of decoding accuracy from a transradial amputee and able-bodied subjects shows *no statistically significant difference* ($p < 0.05$) between these subjects. These results are encouraging for the development of real-time control strategies based on the surface myoelectric signal to control dexterous prosthetic hands.**

***Index Terms*—Electromyography (EMG), myoelectric signals, neural networks, transradial amputee.**

I. Introduction

ADVANCES in upper limb prosthetics have typically lagged behind their lower limb counterparts. It is easy to identify two main reasons behind this "bias": 1) the lower number of upper limb amputations implies a lower market demand for the prostheses [1], [2], making research and development of next-generation prostheses harder to justify; 2) the significantly larger number of *actuated* degrees of freedom (DOFs) involved in controlling the upper extremities requires many more biosignals (even compared to 0, such as in passive prostheses [3]) for a biomimetic control of the prostheses.

Various programs, such as the Defense Advanced Research Projects Agency's (DARPA) Revolutionizing Prosthetics 2009, are helping reverse this bias by placing a new emphasis on upper limb prostheses. For such programs to be successful, two parallel efforts must come to fruition. The first consists of the *development* of an actual prosthetic arm/hand capable of actuating up to 23 DOF depending on the level of amputation. The second requires the ability to *control* these DOFs using different types of biosignals. While acquisition of invasive biosignal from the motor cortex or the peripheral nerve may be suitable for tetraplegics, who have lost their ability to control the entirety of their upper limbs, noninvasive or minimally invasive biosignals such as surface electromyography (sEMG) or intramuscular electromyography from residual limb or reinnervated muscles [4] seem to be the most appropriate solution for upper limb amputees. However, noninvasive or minimally invasive biosignals such as sEMG or intramuscular electromyography from residual limb or reinnervated muscles [4] hold the potential for more near term solutions for upper limb transradial amputees.

Most commercially available upper limb prostheses currently use sEMG signals to control a limited number of DOFs (typically ≤ 2) [5], [6]. However, as efforts in the development of prosthetic hands with high dexterity come to fruition, it will be necessary for the appropriate control systems to follow suit. For example, prosthetic hands like the Shadow dexterous hand, Cyberhand, DLR-HIT hand, or the Fluid hand with more than 16 DOF offer a higher degree of control (see [7] for a summary of many prosthetic and robotic hands under development). While these prosthetic devices are not far from market, no noninvasive-electromyographic (EMG) control system offers the high level of control required by the devices [8].

Various groups have shown that EMG-based control of a prosthesis with greater movement dexterity can be achieved using different combinations of extracted features and classification methods [9]–[13]. As far as the decoding of intramuscular myoelectric signals is concerned, Hargrove *et al*. [14] have recently shown that there was no significant difference when the decoding accuracy of the intramuscular EMG was compared to that of surface-based signals in wrist and grip movements.

However, it is easy to recognize that the greatest level of dexterity corresponds to achieving endpoint control of each individual finger. Jiang *et al*. [15] have shown endpoint control of

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three fingers using features from the wavelet transform of the EMG signals of able-bodied subjects. Our previous work has shown discrete control of each finger in a single able-bodied subject performing individuated finger motion tasks [16].

This study addresses two questions: 1) Is it possible to reliably decode individuated finger and hand movements using surface myoelectric signals from the forearm in able-bodied subjects as well as transradial amputees? And if so, 2) Are there significant differences in overall decoding accuracies between able-bodied subjects and transradial amputees in performing the movements? The significance of the first question will lead to algorithms for the actuation strategy for multifinger hand prosthesis. The second question will lead to improved training and development of the algorithms and understanding of the differences in the muscle signal sources in intact versus transradially amputated arms.

II. Methods

A. Experimental Preparation

Six subjects—four men (A, D–F), two women (B, C), aged 23–26—performed the experiments described in this section. Subjects A, C–F were able-bodied subjects; subject B was a transradial amputee. The results from the decoding accuracy from subject A have been previously documented [16]. All subjects gave their written informed consent and the studies were performed under an approved Johns Hopkins Institutional Review Board (IRB) protocol.

EMG data were acquired using a Compumedics (El Paso, TX) Neuroscan SynAmps^2 64-channel amplifier. The amplifier was connected to a PC for data storage. Auditory and visually animated cues were presented on a computer screen and used for data synchronization. Data were sampled at 2000 Hz and low-pass filtered at 500 Hz.

Before electrode placement, the arms were cleaned with a Nuprep abrasive skin preparation gel from D.O. Weaver & Co. (Aurora, CO). Subject A's arms were also locally shaved to improve signal quality.

An EMG array of bipolar Ag–AgCl electrodes from Myotronics-Noromed (Tukwila, WA) was placed on the subjects' right arm according to the European recommendations for sEMG [17]. This was implemented by dividing the forearm into five distinct levels, as shown in Fig. 1(c), where levels I through V were characterized by 32, 28, 24, 19, and 12 electrodes, respectively. Two Cleartrace LT electrodes from ConMed Corporation (Utica, NY) were used as reference and ground electrodes. The reference electrode was placed on the proximal part of the olecranon and the ground electrode was placed on the clavicle.

B. Experimental Setup

The experiments were first described in [16]. Briefly, the subjects were seated in front of a 50 in screen [see Fig. 1(a)], displaying animations of finger and hand movements. Their primary arm rested the elbow and wrist on two separate compliant surfaces to avoid artifacts generated by contact between electrodes and the resting surfaces, and was bent to form approximately a 90° angle. Twelve different movements were executed

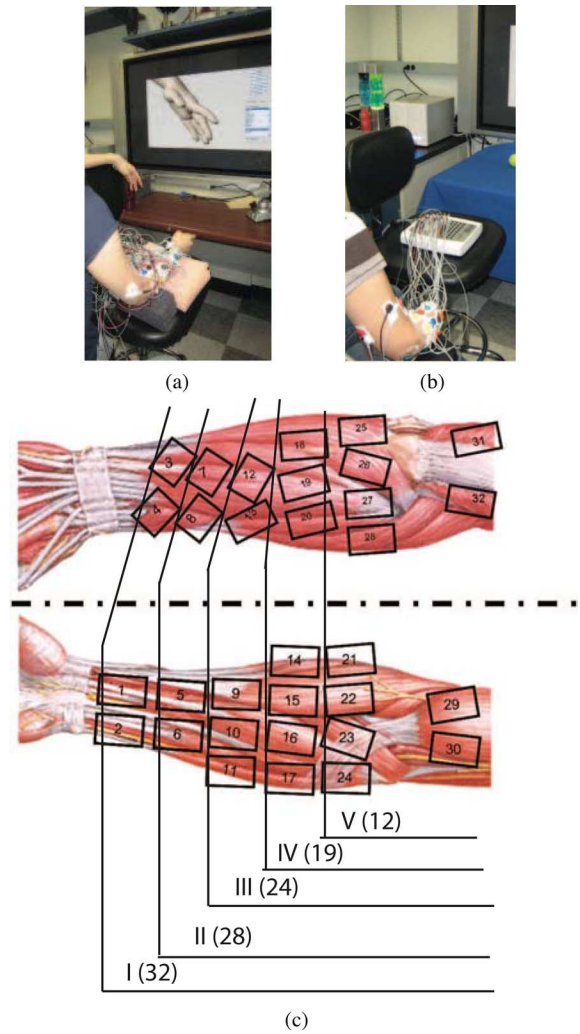

Fig. 1. Experimental setup for (a) subject C and (b) subject B. (c) Approximate location of electrodes as a function of level of amputation: the higher the level, the lower the number of electrodes used. We used level I electrodes for able-bodied subjects and level IV for subject B.

by the subjects in response to a visual animation on the screen and an auditory cue that depicted the movement to be reproduced. The movements consisted of flexions and extensions of all the fingers individually and of the middle, ring, and little finger as a group (MRP group). The movements were denominated $e_1, e_2, \ldots, e_5, e_{345}, f_1, f_2, \ldots, f_{345}$, where e_1 is extension of the thumb, and f_{345} is flexion of the MRP group. Subjects performed each movement 25–30 times (depending on fatigue or other factors) divided in two to three separate blocks of trials. Upon presentation of the (pseudorandom) cues, the subject was asked to flex or extend a specific finger or group of fingers and then keep the position for 3 s until a rest cue was provided. Each individual rest period lasted between 2 and 4 s making each individual trial between 5 and 7 s long. The structure of an individual trial is shown in Fig. 2. The transradial amputee performed the movements in a slightly different manner than the able-bodied subjects by producing the movements simultaneously with both the intact hand/arm and the residual limb, to maximize the reproducibility and reliability of the movements.

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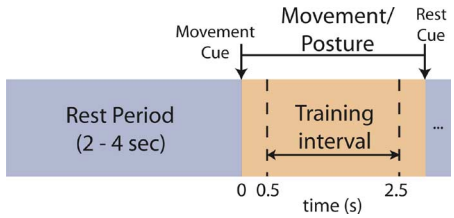

Fig. 2. An individual trial is characterized by a rest period, lasting between 2 and 4 s, between movement events. Movement events are triggered by a cue that is both visual and auditory. Following the cue, the subject performs the movement and holds the contraction until a rest cue is given.

C. Data Processing: Signal Representation

EMG data were bandpass filtered between 10 and 500 Hz as recommended by the Surface Electromyography for the Non-Invasive Assessment of Muscles (SENIAM) protocol [17]. The signals were acquired using a 64 unipolar channel amplifier and subsequently bipolarized by subtracting the channels that correspond to the same electrode.

It has been shown that the success of any pattern recognition system relies heavily on the choice of features used to represent it, even more so than the pattern classifier that is adopted [18]. The simplest approach is to attempt to classify the patterns using the sampled raw myoelectric signals. This, however, results in classification accuracies unsuitable for control purposes [19]. Frequency-based representations are also suited for classification, but the transformations typically involved are often computationally intensive, thus creating a potential obstacle toward real-time decoding of the movements. Additionally, Englehart *et al*. [9] showed that for real-time decoding of four movements of hand and wrist, time-domain features performed better than frequency domain and wavelet packet transforms. To exploit these characteristics, traditional time-domain features were extracted from the preprocessed EMG data.

The chosen features were extracted from sliding time windows of 200 ms in duration every 25 ms. More specifically, with reference to Fig. 2, features were extracted during the 2 s time interval that begins 500 ms following presentation of the cue. This choice allows accounting for delays in reaction time between cue presentation and finger action. Fig. 3(a) shows the raw EMG signal from a specific electrode. Four features were extracted and compared as a result of the subsequent classification. They were chosen based on the underlying information they contain for myoelectric signals and on their ease of implementation in embedded microcontrollers.

1) *Mean of the absolute value*:

$$\overline{X} = \sum_{i=1}^{N} |x_i| \tag{1}$$

where x_i are the individual EMG data, N is the number of datapoints in each feature, given by the sampling frequency multiplied by window size (i.e., 2000 Hz $\times$ 0.2 s $=$ 400). This feature displays a large increase in value at onset and maintains fairly high values during contraction. This feature is presented in Fig. 3(b).

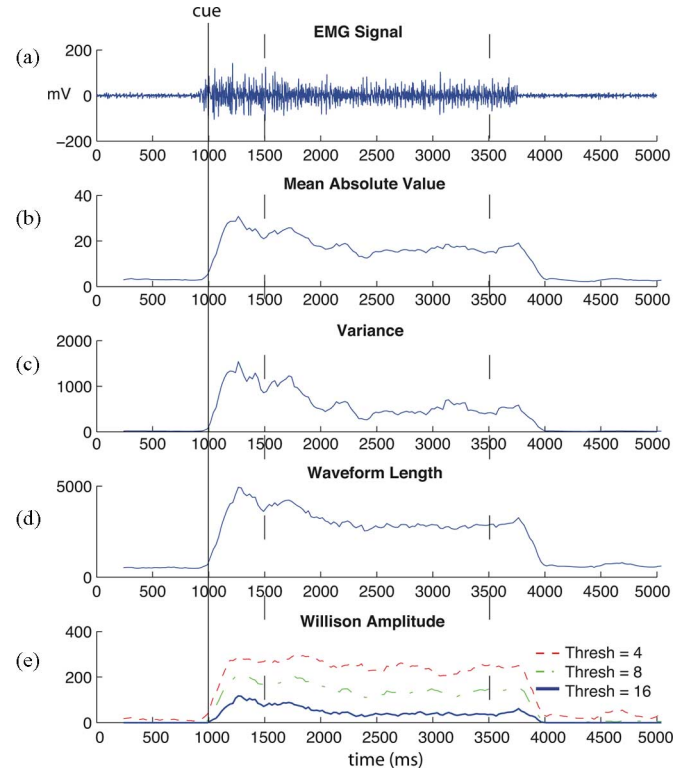

Fig. 3. (a) Raw EMG for a given trial recorded by a specific electrode and relative features extracted from it. (b) Mean of the absolute value. (c) Variance. (d) Waveform length. (e) Willison amplitude for different threshold values. The solid vertical line indicates the time at which the cue was presented, whereas the dashed vertical lines represent the actual data upon which the features are calculated and used in training.

2) *Variance*:

$$\sigma^2 = \frac{1}{N-1} \sum_{i=1}^{N} x_i^2 \tag{2}$$

this feature is a representation of the EMG signal power, useful in determination of onset and contraction. It is shown in Fig. 3(c).

3) *Waveform length (WL)*:

$$WL = \sum_{i=1}^{N} |x_i - x_{i-1}|. \tag{3}$$

The waveform length of the signal provides indicators for signal amplitude and frequency and is shown in Fig. 3(d).

4) *Willison amplitude (W)*:

$$W = \sum_{i=1}^{N} f(|x_i - x_{i-1}|) \tag{4}$$

where

$$f(x) = \begin{cases} 1, & \text{if } x > \text{threshold} \\ 0, & \text{otherwise.} \end{cases} \tag{5}$$

The Willison amplitude for a given interval represents the number of times the magnitude of the difference for any two consecutive points exceeds a certain threshold. Its value correlates to muscle contraction levels [8]. An

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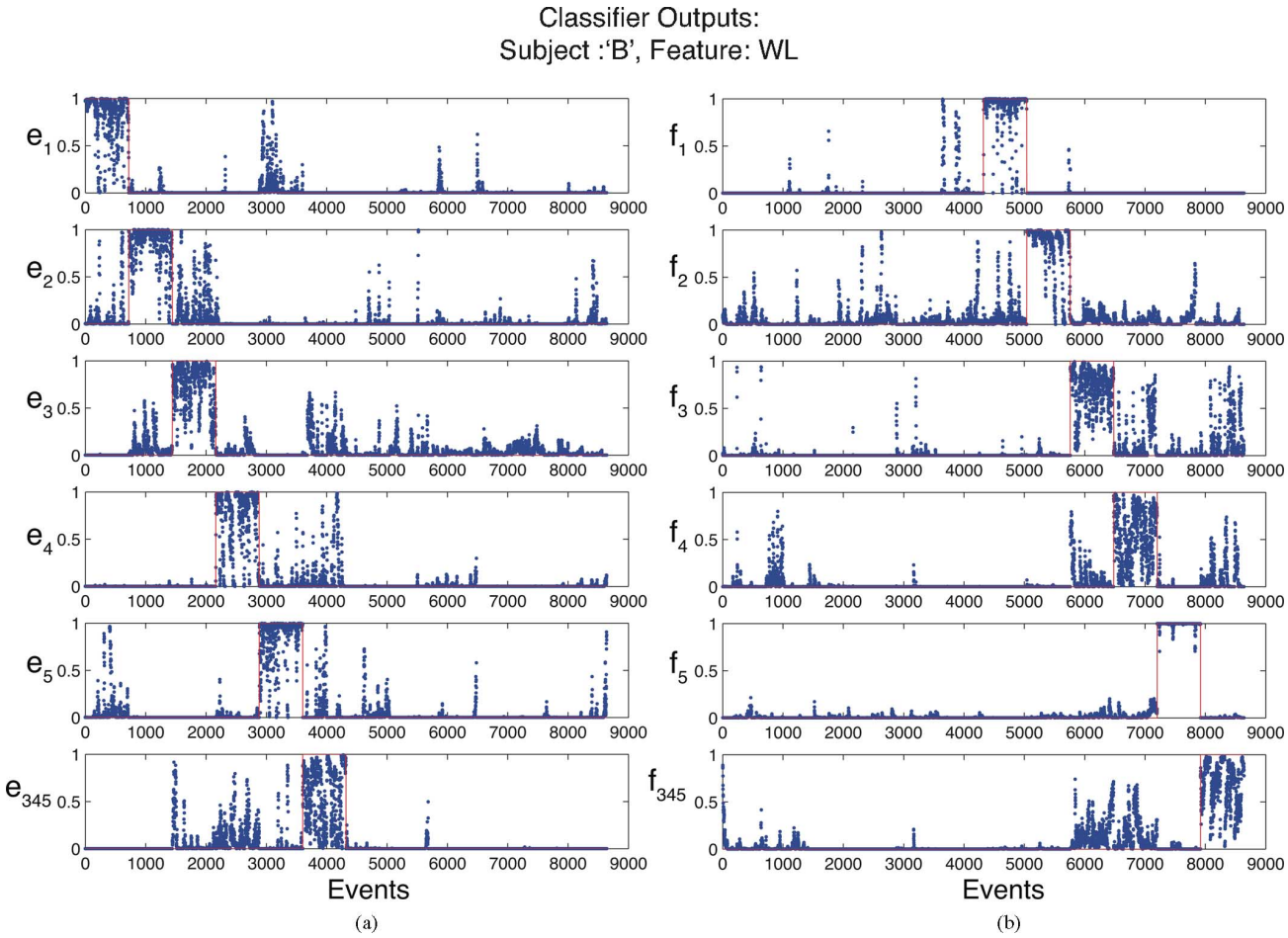

Fig. 4 (a) Classifier outputs (dots) and actual movement (lines) for the 12 movements performed by the transradial amputee. Each movement represents one of the classifier's 12 output neurons. The output label that is ultimately selected for a given time point is given to the neuron with the highest value at that time point. (b) Average decoding accuracy for each movement for each subject as a function of time (within the training/testing time interval). As can be seen, very accurate decoding is achieved throughout the time interval, simplifying the transition to a real-time prosthesis.

example of this feature is shown in Fig. 3(e) for three different values of the threshold.

The 200-ms duration of the analysis window was chosen to target real-time classification, by minimizing the delay between performed and decoded action, while still providing sufficient data for a valid feature to be extracted and accurately decoded. Finally, each window was shifted by 25 ms, thereby creating overlapping adjacent windows (herein called events), to allow reasonable continuity in the extracted feature.

D. Classification

To further help the classifier distinguish the different movements, the input space (extracted features) was decorrelated through transformation in a principal component space (performing a discrete Karhunen–Loeve transform). All the extracted components were retained, since the size of the training dataset did not warrant a reduction of dimensionality. Feedforward multilayer perceptrons (MLPs) with varying numbers of hidden layer neurons (empirically chosen to be between 0.5 and 4 times the dimension of the input space) were used for classification, and 12 output neurons, i.e., one neuron per movement type. The use of a continuous transfer function for the hidden layer neurons (tan-sigmoid) and a logistic function for the output layer, allows us to assign a probability, $P\{M_i\}$, where $i = 1, \ldots, 12$, for each movement (or output neuron). The output neuron with the *maximum value* (equivalent to the one with the highest probability) is selected as the classifier's output. The MLPs were trained using MATLAB's scaled conjugate gradient descent algorithm in combination with early validation to improve generalization. More details on this approach can be found in [20]. The training, validation, and testing sets were kept in separate but not fixed ratios of 0.50 (training)/0.20–0.25 (validation)/0.30–0.25 (testing).

Fig. 4 shows (in dots) the values of the 12 output neurons corresponding to the movements (f_1–e_{345}) from the waveform length extracted from the data recorded from a transradial amputee at 25 ms time intervals (corresponding to the time between two consecutive windows). The continuous line shows the intended movement. At each event (25 ms, i.e., the time between adjacent windows), the classifier chooses as output the neuron (i.e. movement type) with the highest value.

Using these decoded accuracies for each individual movement—as repeated measures of the same treatment in

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TABLE I
SUMMARY OF DECODING ACCURACY FOR ALL SUBJECTS (IN PERCENTAGE)

Subject	No. elec. (Level)	Feat.	Acc.	Subject	No. elec. (Level)	Feat.	Acc.
A	32 (I)	MAV	98.8±1.1	D	32 (I)	MAV	92.5±6.4
		Var	98.8±0.9			Var	90.9±8.4
		WA	98.6±2.5			WA	84.9±4.6
		WL	99.7±0.3			WL	95.0±6.1
	19 (IV)	MAV	98.3±2.1		19 (IV)	MAV	88.8±9.4
		Var	98.1±2.3			Var	87.8±10.0
		WA	98.0±2.7			WA	80.0±13.0
		WL	99.1±1.3			WL	94.3±5.7
				E	32 (I)	MAV	92.1±5.8
						Var	86.9±8.2
						WA	91.4±6.0
						WL	93.5±3.6
B	19 (IV)	MAV	82.3±15.7		19 (IV)	MAV	88.9±7.4
		Var	83.8±13.1			Var	81.6±16.7
		WA	83.3±14.8			WA	87.5±8.9
		WL	87.8±12.3			WL	91.5±6.4
C	32 (I)	MAV	88.8±8.0	F	32 (I)	MAV	89.2±7.5
		Var	87.7±8.0			Var	86.4±7.1
		WA	86.9±9.8			WA	89.4±6.9
		WL	93.6±6.2			WL	89.6±5.1
	19 (IV)	MAV	84.4±10.9		19 (IV)	MAV	89.2±5.9
		Var	84.8±11.2			Var	85.6±6.5
		WA	86.9±11.8			WA	89.2±6.2
		WL	92.7±7.6			WL	89.1±5.7

Kruskal–Wallis (KW) tests[1]—we test two hypotheses: 1) there is no significant difference in decoding accuracy of individuated and combined MRP finger movements in able-bodied subjects whether level I or level IV electrodes are used to decode; 2) there is no significant difference in decoding accuracy of individuated and combined MRP finger movements between able-bodied subjects using level I electrodes and a transradial amputee of level IV amputation.

The outcome of these tests ($p < 0.05$) should be interpreted as follows. If a statistically significant difference in decoding accuracy between different electrode levels is found [cf. Fig. 1(b)], then the accuracies processed in transradial amputees will be a function of the length of the residual limb. If there is no statistically significant difference between the two, then it is reasonable to assume that there should not be a significant difference in decoding accuracies between able-bodied subjects and transradial amputees, and the outcome of the second test should show this.

III. RESULTS

The classifier's overall accuracy (averaged across the 12 movement types) in decoding the individuated and combined finger movements of the six subjects is summarized in Table I. The table presents two types of results:

1) decoding accuracy for each subject using a number of electrodes dependent on the subjects' arm (which corresponds to level I from Fig. 1(c) for the able-bodied subjects and level IV for the transradial amputee);
2) decoding accuracy for the able-bodied subjects using the same number of electrodes (and in the same positions) as for the transradial amputee (level IV).

[1]KW tests were used instead of a parametric analysis of variance (ANOVA) due to the nonnormal distribution of the accuracy data.

Fig. 5 shows the suitability of the system for real-time decoding, in which the accuracy of each testing window in the 2 s interval (as depicted in Fig. 2) for each movement is averaged between all the test trials, for the six subjects.

Fig. 6 shows the confusion matrices for the six subjects, to better visualize the differences in decoding accuracies among the different movements.

KW tests performed on the decoding accuracies of able-bodied subjects show that there is no statistically significant difference ($p < 0.05$) in decoding accuracy using 32 or 19 electrodes (i.e., levels I or IV).

KW tests were then performed on the six subjects, using accuracy values obtained using the *maximum* number of electrodes for each subject (32 for the able bodied subjects, 19 for the transradial amputee). The result of this test shows a statistically significant difference ($p < 0.05$) between subject A and subjects B–F, but no difference between subjects B—F. The Section IV elaborates on this observation.

IV. DISCUSSION

This study shows that it is possible to decode individual and combined MRP finger movements in a transradial amputee with high accuracy. Furthermore, we show that there is no significant difference in decoding accuracy between most able-bodied subjects and a transradial amputee performing finger movement tasks.

A. On Statistical Tests and Confusion Matrices

The high decoding accuracies by all the subjects notwithstanding, the statistically significant difference between subject A and all other subjects (B–F, $p < 0.05$) may be attributed to the fact that subject A had performed this series of experiments a number of times, prior to this data collection. Therefore, there may be a distinction to be made between "trained" and

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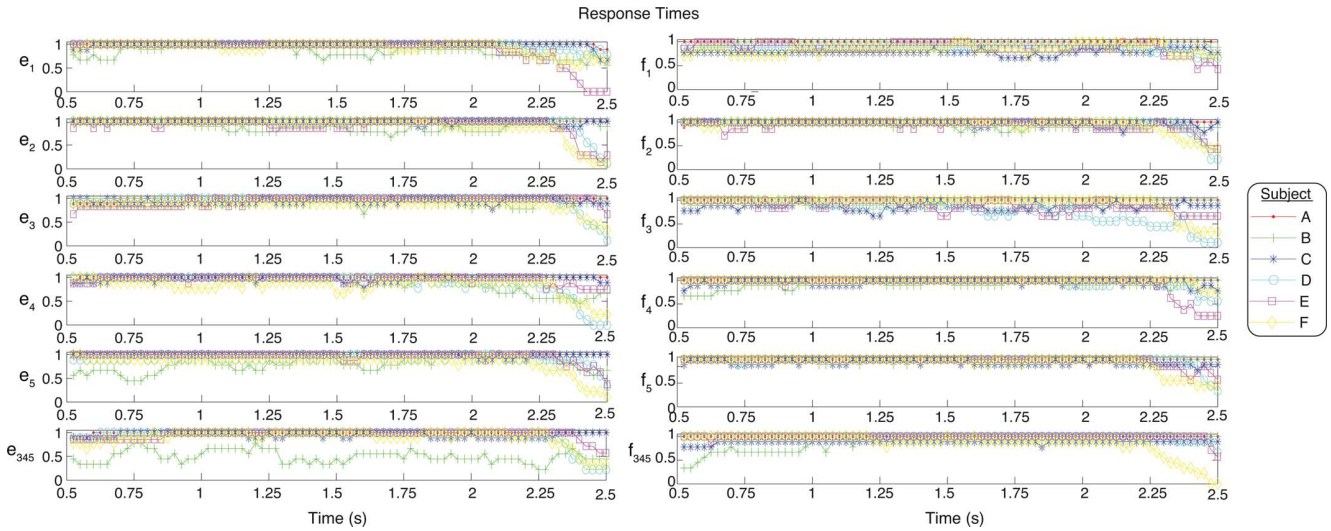

Fig. 5. Average decoding accuracy for each movement for each subject as a function of time (within the training/testing time interval). As can be seen, very accurate decoding is achieved throughout the time interval, simplifying the transition to a real-time prosthesis.

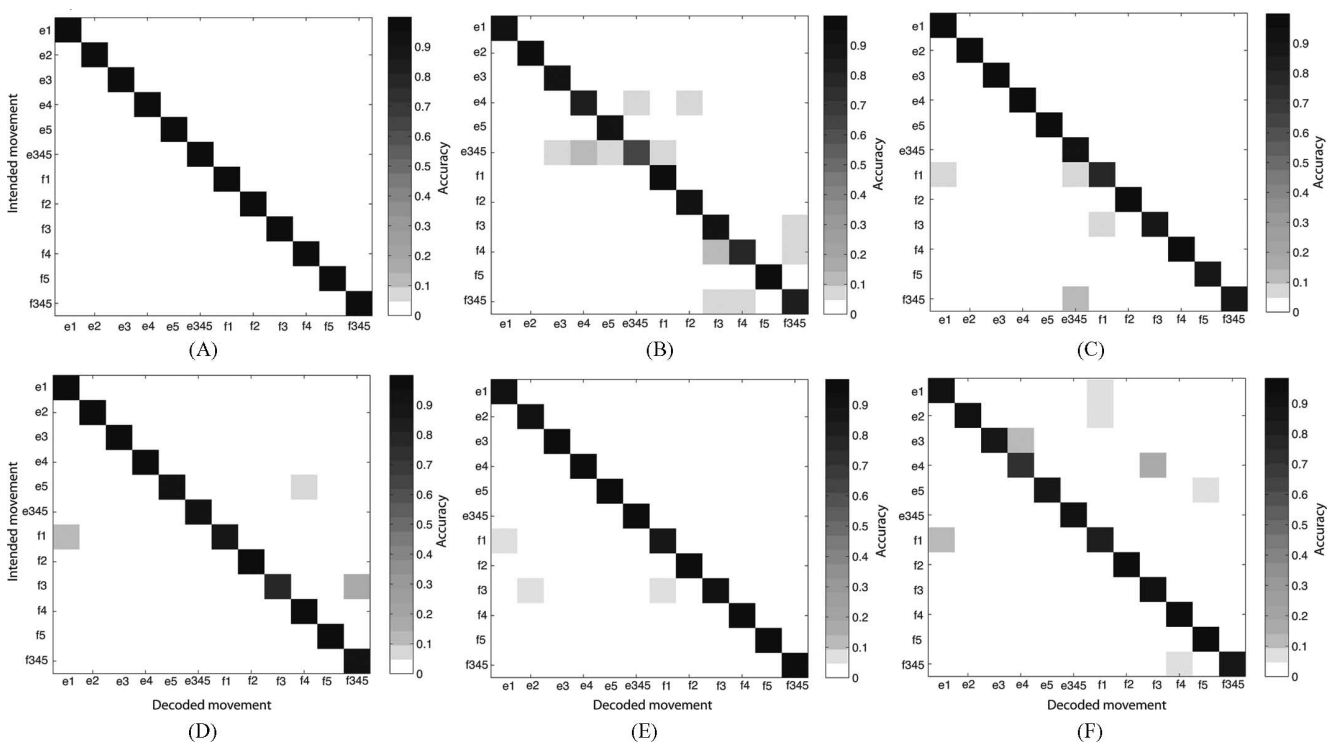

Fig. 6. Confusion matrices of the six subjects (A–F). As can be seen, the accuracy of the classifier output for the MRP group movement of the transradial amputee is somewhat worse (75% decoding accuracy) than the accuracy of the individual fingers. Since the confusion arises between the MRP-group's output neuron and the output neurons of the individual movements that make up the MRP-group movement, the lower decoding accuracy is likely due to the choice of the classifier output. The confusion matrices are obtained using 32 electrodes for the able-bodied subjects and 19 electrodes for the transradial amputee.

"untrained" subjects. For example, in their study on the effects of training on movements of the affected hand of hemiparetic patients, Butefisch *et al.* [21] found that repetitive practice of hand and finger movements resulted in significantly improved motor performance.

However, the confusion matrices shown in Fig. 6 present us with a potentially different interpretation. As can be seen, subjects C–F present confusion between antagonist movements (for example, the classifier output for subject C confuses movements f_{345} with e_{345}, and f_1 with e_1; similarly, subject D's classifier output confuses f_1 with e_1). On the other hand, the majority of the confusion in subject B is in discerning the combined finger movements (e_{345} and f_{345}) from the individual movements that constitute the group movement. A tracking device that ensures that a subject performs the correct movement is important to understand wherein the error lies, whether in the subject performing an incorrect movement or in the classifier itself. This, of course, being achievable only with able-bodied subjects.

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Interestingly, if we disregard the combined finger accuracies, the average accuracy in decoding individual finger movements alone performed by the transradial amputee is 92.3%.

Moreover, the misclassification of the group movement in subject B implies that differentiating between combined finger movements and the movements that comprise them may not be natural, and may require more training. Future experiments that include all possible combinations of individual and combined finger movements will provide a better understanding of this type of confusion.

The limited number of subjects used in these experiments makes it difficult to generalize the results to the general population, both able-bodied and amputees. However, it is likely that the results from a broader population will not significantly diverge from these results because of the large number of electrodes used to record the myoelectric signals that provides ample data that enable to extract relevant information and classify correctly. Jiang *et al*. [15] have shown that as few as four pairs of electrodes were sufficient to decode extension and flexion of three fingers with reasonable accuracy (70%–80%) in four able-bodied subjects. Huang *et al*. [22] used a three-channel EMG signal to decode six different grips (eight movements total).

B. On Recording Space

The large numbers of electrodes used in these experiments should not be considered a hindrance toward embedding in an actual prosthesis. Discussions with prostheses makers (such as with Dr. Charles Dankmeyer, Dankmeyer Orthotics and Prosthetics, Inc.) have brought us to the conclusion that there is no practical limit to the number of electrodes that can be instantiated in a prostheses. Additionally, they allow a practical demonstration of the potential for decoding the movement of each individual fingers in able-bodied subjects and transradial amputees alike.

C. On Suitability for Real-Time Prostheses

The myoelectric control system described here satisfies three crucial conditions for it to be applicable to real-time decoding [9]: *accuracy* is necessary to fulfill the user's intent; a *direct mapping* of the controls allows the user to require minimal training to operate the prostheses; and finally a *low response time* to make delays in the control appear insignificant to the user (typically less than 300 ms). These three conditions are shown to be satisfied in Fig. 5, which depicts the average accuracy for each subject for each movement and for each 25 ms shift in the time-window. It must be noted, however, that the accuracy reported here is based on the steady-state myoelectric signal, which is associated with a constant contraction. This was done because the ultimate goal of this work was to examine the possibility of decoding individual finger movements in a transradial amputee. In an able-bodied subject, in fact, it would have been possible to use a tracking system for the hand, such as a Cyberglove [23], to identify when the movement was actually performed. In unilateral transradial amputees, however, a good proxy for tracking movement using such a device is to collect position data from the intact hand, while recording the EMG on the residual limb, similarly to the way that the experiment was performed by the transradial amputee. However, in the absence of such a device, it is only possible to rely on the instant of cue presentation and use it as a proxy for movement.

Nonetheless, the extracted features in Fig. 3 show that decoding the transient as well as the steady-states, as done in [19], would have been feasible using fuzzy decision boundaries around the time of cue presentation. Englehart *et al*. [9], [24] use a wavelet-based approach as opposed to a time-domain approach to decode movements in real time foregoing the need for a rest state in between events. This approach requires more complex hardware than typical time-domain strategies (ultimately achievable on state-of-the-art microcontrollers or DSP-based microprocessors), but the high decoding accuracies obtained using features in the time domain do not warrant a more complex approach.

The long-term goal of this study is to achieve real-time decoding of individual finger movements in transradial amputees and use it to control modern multifinger hand prostheses. The results presented in this paper provide the signal processing approach required to decode the individuated finger movements.

V. Conclusion

This study presents the first work on decoding individual finger movements performed by a transradial amputee using sEMG electrodes placed around the residual limb. We have shown that a high level of decoding accuracy ($\sim 90\%$) can be achieved using a neural network classifier trained on a waveform length feature. Moreover, there appears to be no significant difference in accuracy results between able-bodied subjects and transradial amputees, in absence of training.

References

[1] D. S. Childress, "Historical aspects of powered-limb prostheses," *Clini. Prosthet. Orthot.*, vol. 9, no. 1, pp. 2–13, 1985.

[2] D. J. Atkins, D. C. Y. Heard, and W. H. Donovan, "Epidemiologic overview of individuals with upper-limb loss and their reported research priorities," *J. Prosthet. Orthot.*, vol. 8, no. 1, pp. 2–11, 1996.

[3] M. Hood, "Running against the wind," *IEEE Spectr.*, vol. 42, no. 6, pp. 13–14, Jun. 2005.

[4] T. Kuiken, "Targeted reinnervation for improved prosthetic function," *Phys. Med. Rehabil. Clinics North Amer.*, vol. 17, pp. 1–13, 1985.

[5] V. S. Nelson, K. M. Flood, P. R. Bryant, M. E. Huang, P. F. Pasquina, and T. L. Roberts, "Limb deficiency and prosthetic management. 1. decision making in prosthetic prescription and management," *Arch. Phys. Med. Rehabil.*, vol. 87, pp. S3–S9, 2006.

[6] R. A. Roeschlein and E. Domholdt, "Factors related to successful upper extremity prosthetic use," *Prosthet. Orthot. Int.*, vol. 13, pp. 14–18, 1989.

[7] [Online]. Available: http://www.shadowrobot.com/hand

[8] M. Zecca, S. Micera, M. Carrozza, and P. Dario, "Control of multifunctional prosthetic hands by processing the electromyographic signal," *Crit. Rev. Biomed. Eng.*, vol. 30, pp. 459–485, 2002.

[9] K. Englehart and B. Hudgins, "A robust, real time control scheme for multifunction myoelectric control," *IEEE Trans. Biomed. Eng.*, vol. 50, no. 7, pp. 848–854, Jul. 2003.

[10] K. R. Wheeler, M. H. Chang, and K. H. Knuth, "Gesture-based control and EMG decomposition," *IEEE Trans. Syst., Man, Cybern. A Syst., Humans*, vol. 36, no. 4, pp. 504–513, Jul. 2006.

[11] K. A. Farry, I. D. Walker, and R. G. Baraniuk, "Myoelectric teleoperation of a complex robotic hand," in *Proc. IEEE Int. Conf. Robot. Autom.*, 1996, vol. 12, no. 5, pp. 775–788.

[12] G. Wang, Z. Wang, W. Chen, and J. Zhuang, "Classification of surface EMG signals using optimal wavelet packet method based on davies-bouldin criterion," *Med. Biol. Eng. Comput.*, vol. 44, pp. 865–872, 2006.

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[13] D. Peleg, E. Braiman, E. Yom-Tov, and G. F. Inbar, "Classification of finger activation for use in a robotic prosthesis arm," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 10, no. 4, pp. 290–293, Dec. 2002.
[14] L. J. Hargrove, K. Englehart, and B. Hudgins, "A comparison of surface and intramuscular myoelectric signal classification," *IEEE Trans. Biomed. Eng.*, vol. 54, no. 5, pp. 847–853, May 2007.
[15] M. W. Jiang, R. C. Wang, J. Z. Wang, and D. W. Jin, "A method of recognizing finger motion using wavelet transform of surface EMG signal," in *Proc. IEEE Conf. Eng. Med. Biol. Soc.*, 2005, pp. 2672–2674.
[16] F. Tenore, A. Ramos, A. Fahmy, S. Acharya, R. Etienne-Cummings, and N. V. Thakor, "Towards the control of individual fingers of a prosthetic hand using surface EMG signals," in *Proc. IEEE Eng. Med. Biol. Soc. Conf.*, 2007, pp. 6145–6148.
[17] J. H. Blok and D. F. Stegeman, "Simulated bipolar sEMG characteristics," in *SENIAM 5: The State of the Art on Sensors and Sensor Placement Procedures for Surface Electromyography: A Proposal for Sensor Placement Procedures*, H. J. Hermens and B. Freriks, Eds., Enschede, The Netherlands: Roessingh Research and Development, 1997.
[18] K. Englehart, B. Hudgins, P. A. Parker, and M. Stevenson, "Classification of the myoelectric signal using time-frequency based representations," *Med. Eng. Phys.*, vol. 21, pp. 431–438, 1999.
[19] B. Hudgins and P. Parker, "A new strategy for multifunction myoelectric control," *Trans. Biomed. Eng.*, vol. 40, no. 1, pp. 82–94, 1993.
[20] V. Aggarwal, S. Acharya, F. Tenore, H. C. Shin, R. Etienne-Cummings, M. H. Schieber, and N. V. Thakor, "Asynchronous decoding of dexterous finger movements using M1 neurons," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 16, no. 1, pp. 3–14, Feb. 2008.
[21] C. Butefisch, H. Hummelsheim, P. Denzler, and K. H. Mauritz, "Repetitive training of isolated movements improves the outcome of motor rehabilitation of the centrally paretic hand," *J. Neurol. Sci.*, vol. 130, pp. 59–68, 1995.
[22] H. P. Huang and C. Y. Chen, "Development of a myoelectric discrimination system for a multi-degree prosthetic hand," in *Proc. IEEE Int. Conf. Robot. Autom.*, 1999, pp. 2392–2397.
[23] [Online]. Available: http://www.immersion.com/3d/products/cyberglove.php
[24] K. Englehart and B. Hudgins, "A wavelet-based continuous classification scheme for multifunction myoelectric control," *IEEE Trans. Biomed. Eng.*, vol. 48, no. 3, pp. 302–311, Mar. 2001.

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